

Attack on Finances (AoF)

You are a cybersecurity analyst for a large financial institution. One day, you receive an urgent message from your boss informing you that the company's payment processing system has been hacked. The hackers have stolen millions of dollars from the company's accounts, and they're threatening to leak sensitive customer information if their demands aren't met. Your boss tells you that the company's IT team has identified the source of the attack as a group of hackers based in Eastern Europe. They've traced the hackers' location to Bucharest in Romania, but they don't have any other information about the group or their motives. Your job is to work with the IT team to set up a network topology that will help you prevent them from accessing any more sensitive data. You'll need to set up a network that covers all of the company's branches across the world, as well as a secure connection to the Romanian clients. The company's headquarters is in Dhaka, and they have branches in California, Lisbon, Paris and Tokyo.

The distance between each branch and the number of computer devices at each branch are listed below. The numbers in brackets [] specify the number of devices in the city and the values in the table specify the distance (in kilometers) between branches. :

City Name	Dhaka	California	Lisbon	Paris	Bucharest	Tokyo
Dhaka [44AB]	0	11	-	56	72	-
California [300]	11	0	200	-	-	152
Lisbon [200]	-	200	0	-	-	-
Paris [2450]	56	-	-	0	-	-
Bucharest [756]	72	-	-	-	0	-
Tokyo [10XY]	-	152	-	-	-	0

Here, **AB** is the last two digits of STUDENT ID of group member 1 (any)

Here, **XY** are the last two digits of STUDENT ID of group member 2 (any)

Instructions :

- Draw the given topology, where every city mentioned can be denoted as Routers.
- Choose an appropriate network address (NA) for the whole network and create subnets to assign to each of the places according to their requirements.
- **California** and **Tokyo** will also have two separate web servers. A DNS server will be located in **California**. If anyone types the URL “www.california.gov” the web server located at **California** will handle the query and the user will see a webpage that says “**Welcome to Hotel California!**”. Similarly, the web server located in **Tokyo** will handle requests for “www.japan.com” and return “**I love Anime!**” when visited through the URL.
- **Dhaka** and **Bucharest** are the most important cities; for security, they will use static addressing while the other branches’ IP addresses will be assigned using DHCP and handled by their network’s DHCP server.
- Showing at least 2 end devices per network is good enough to represent the whole population. **Paris** has laptops and printers.
- **Lisbon** has a Secure server that only denies access from one PC named “**Vulnerable PC**”, which is located in **Paris**.
- As the company headquarters, the **Dhaka office** has 3 different vendors working within the building (Duber,Uthao,Kharaz) using their computers, who do not want their group of devices to communicate with each other, but still communicate within themselves.
- You must remember that the default routes cannot be used while exchanging any packets. Data will be delivered using static or dynamic routes only.
- Configure half of the network to be routed dynamically.
- You need to be able to ping each end-device (exceptions given above) to another after all the setups are complete.
- **Bonus Task:** **Paris** and **Bucharest** will be communicating a lot, which is why they will require **Email servers** to be set up so that they can exchange mail among themselves. Make sure the email configurations are all set up for sending mail, receiving mail, and replying to mail.